

Which is an environmental benefit of using wind turbines for electricity generation?

[1 mark]

A Greenhouse gases are not produced during the manufacture of the wind turbines.

☐

B Plant or animal habitats are not damaged by the use of wind turbines.

☐

C Fossil fuels are not used by the wind turbines during electricity production.

☐

D Wind turbines do not produce any noise pollution.

☐

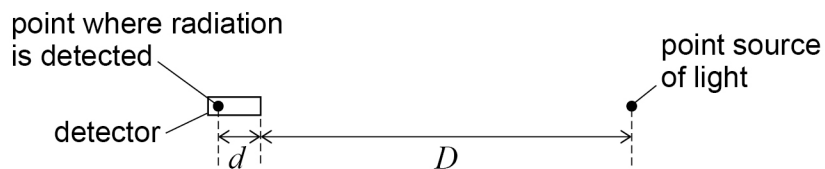
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3 4

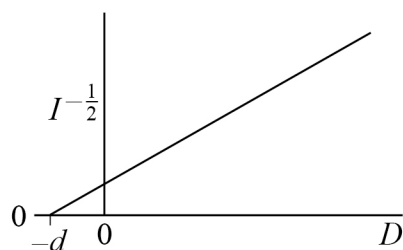
A student measures the intensity I of the radiation from a point source of light. The point source is a distance D from the front of the detector. The radiation is detected at an additional distance d inside the detector.



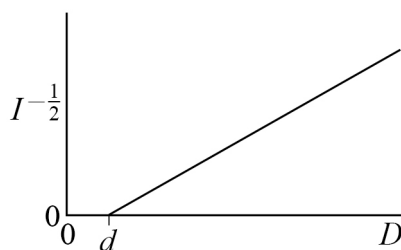
Which graph shows the expected relationship between I and D ?

[1 mark]

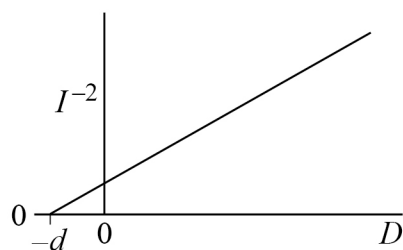
A



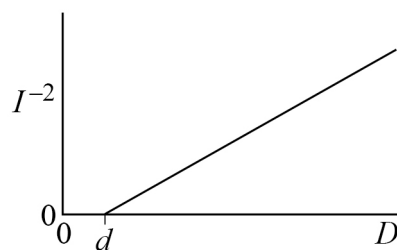
B



C



D



A



B



C



D



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Hydroelectric power is used for electricity generation. Energy for the process comes from the Sun.

0	3
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1

Describe the energy transfers in this process, starting with energy transfers in the Sun.

[4 marks]

[illegible]

03.2

Compare the environmental impacts of hydroelectric power with the environmental impacts of power generation using nuclear fission.

[3 marks]

7

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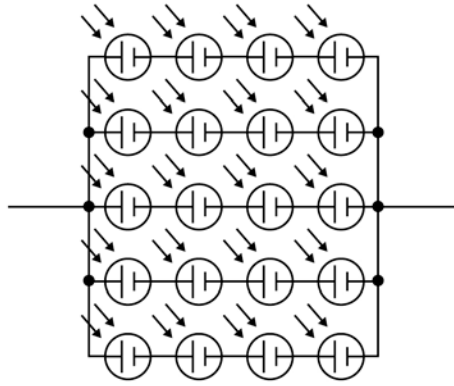


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3

4

20 solar cells are connected together in the solar array shown. Each cell has an output of 13 V and 3.0 A.



What is the potential difference and the current output of the array?

[1 mark]

	Potential difference / V	Current output / A	
A	52	12	☐
B	52	15	☐
C	65	12	☐
D	65	15	☐



Do not write	Do not write
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3 5

An ideal wind turbine has an output power P when the speed of the wind is v . The blade length is r .

A second ideal wind turbine operates when the wind speed is $\frac{v}{2}$ and has a blade length of $3r$.

What is the output power of the second wind turbine?

[1 mark]

A $\frac{4}{27}P$

☐

B $\frac{8}{9}P$

☐

C $\frac{9}{8}P$

☐

D $\frac{27}{4}P$

☐

[illegible]

[illegible]

[illegible]

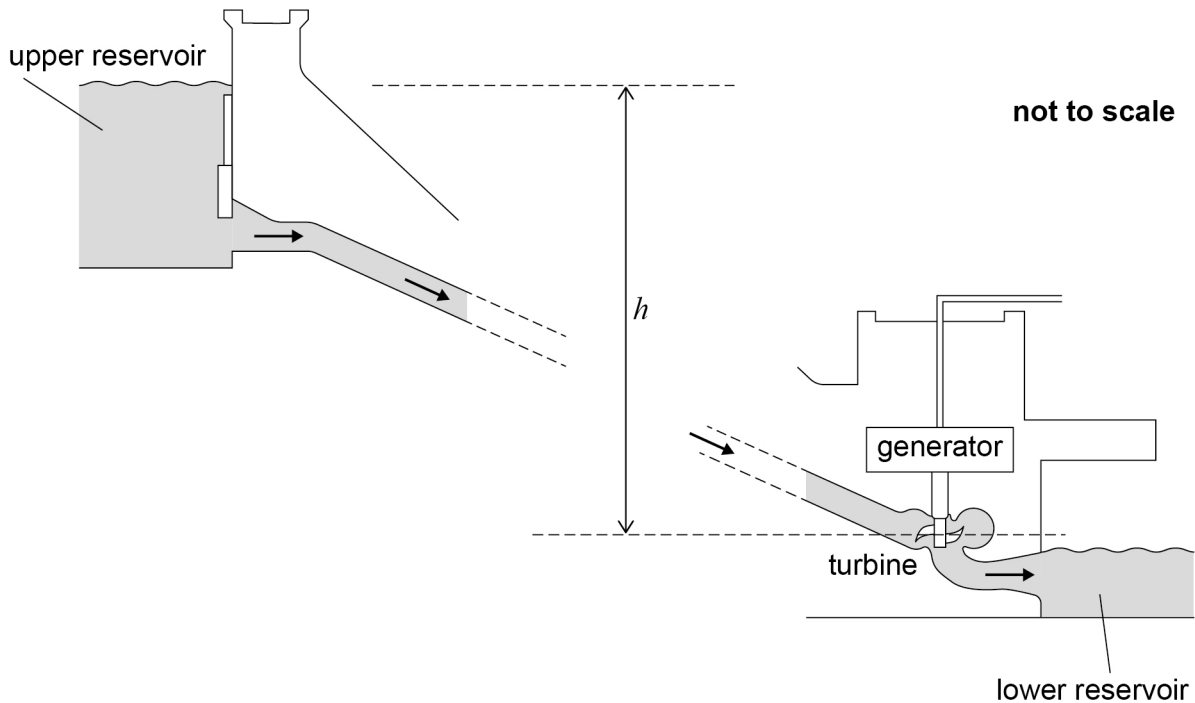
0 3

There are two stages to the operating cycle of a pumped storage system (PSS):

- a generating stage when water falls from the upper to the lower reservoir
- a pumping stage when water is returned to the upper reservoir.

Figure 4 shows a PSS being used to generate electrical power.
The water falls a vertical distance h from the upper reservoir to the turbine.

Figure 4



0 3 . 1

Show that the theoretical maximum power $P_{\max}$ available from the water during the generating stage is given by

$$P_{\max} = \rho g h \frac{\Delta V}{\Delta t}$$

where:

- ρ is the density of the water
- $\frac{\Delta V}{\Delta t}$ is the volume of water flowing through the turbine per unit time.

[1 mark]

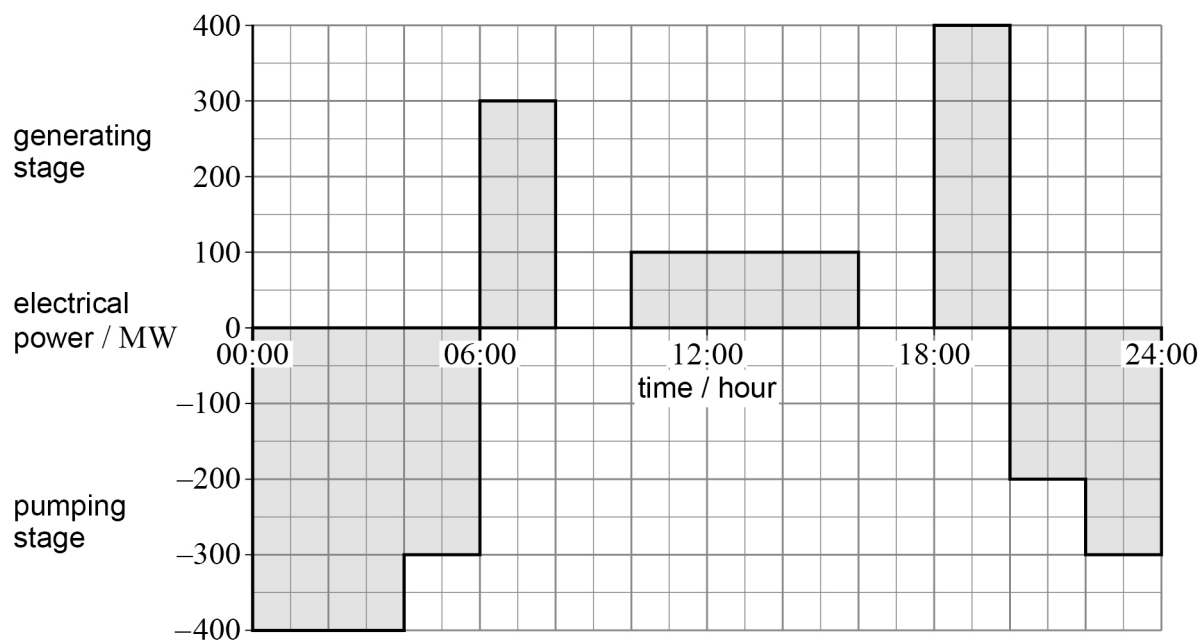




Figure 5 shows the variation in the transfer of electrical power for the PSS with time during a 24-hour period.

In the generating stage, values of power are positive.
In the pumping stage, values of power are negative.

Figure 5



0 3 . 4

Determine, in J, the **net** electrical energy output of the PSS for the 24-hour period shown in **Figure 5**.

[3 marks]

net electrical energy output = _____ J



0 3 . 5

Explain why the overall efficiency of the PSS, for the 24-hour period, is significantly less than the generating efficiency of 90%.

[1 mark]

0 3 . 6

Explain **two** reasons why pumped storage systems are useful.

[2 marks]

1

2

10

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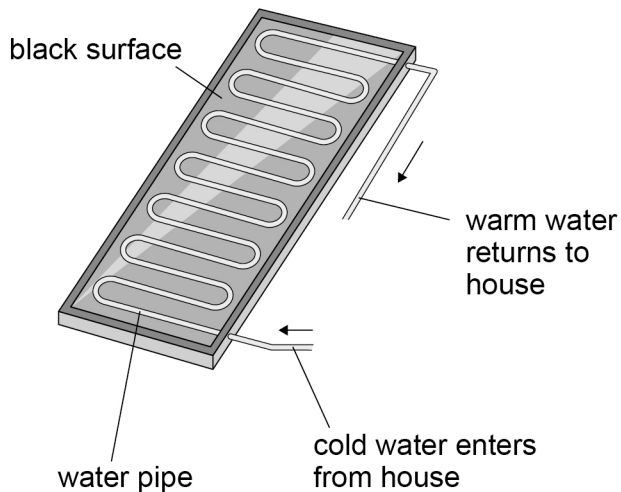


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0 4

Figure 6 shows a thermal solar panel to be used on the roof of a house. Water is heated by solar radiation as it flows through a pipe in the solar panel.

Figure 6



0 4 . 1

Not all of the solar radiation incident on the Earth's upper atmosphere reaches the Earth's surface.

The proportion R of solar radiation incident on the Earth's surface is given by:

$$R = \frac{\text{solar power per m}^2 \text{ incident on the Earth's surface}}{\text{solar power per m}^2 \text{ incident on the Earth's upper atmosphere}}$$

Solar radiation is incident normally on the surface of the solar panel with a power of $2.2 \times 10^3 \text{ W}$.

Calculate R .

total power output of Sun = $3.85 \times 10^{26} \text{ W}$
 distance between Earth and Sun = $1.50 \times 10^{11} \text{ m}$
 area of the upper surface of the panel = 5.5 m^2

[3 marks]

$R =$ _____



0 4 . 2

The rate of flow of water through the panel is 45 g s^{-1} .
The water enters the panel at a temperature of 15°C and absorbs energy from the panel at a rate of 1.7 kW .

Calculate the temperature of the water as it leaves the panel.

specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$

[2 marks]

temperature = _____ $^\circ\text{C}$

5

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ANSWER IN THE SPACES PROVIDED**



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0 4

The energy sources on a small island consist of a combination of a wind farm and a pumped storage system (PSS).

0 4 . 1

Describe how this combination can provide a continuous supply of electrical energy.

[2 marks]

Question 4 continues on the next page



0 4 . 2

The wind farm has several wind turbine generators (WTGs).
One WTG has a rotor diameter of 71 m and an efficiency of 28%.

Calculate the minimum wind speed that could produce a power output of 2.3 MW
from the WTG.

density of air = 1.3 kg m^{-3}

[3 marks]

minimum wind speed = _____ m s^{-1}

0 4 . 3

In the PSS, water flows from an upper reservoir through a pipe to a water turbine.
The water falls a vertical distance of 650 m before striking the turbine.

Calculate the maximum speed of the water striking the turbine.

Assume that there are negligible energy losses as the water flows through the pipe.

[2 marks]

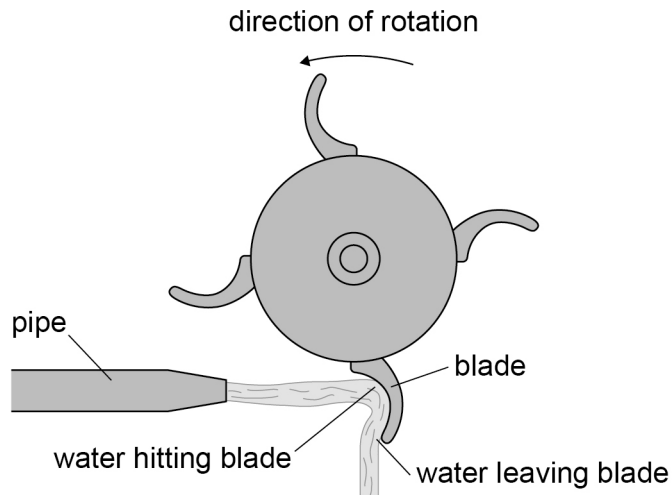
maximum speed of water = _____ m s^{-1}



0 4 . 4

Figure 6 shows water hitting a turbine blade. The water is travelling horizontally at a high speed when it hits the blade. The water leaves the blade in a different direction and with a much lower speed.

Figure 6



Explain how this process produces a torque on the turbine.

[4 marks]



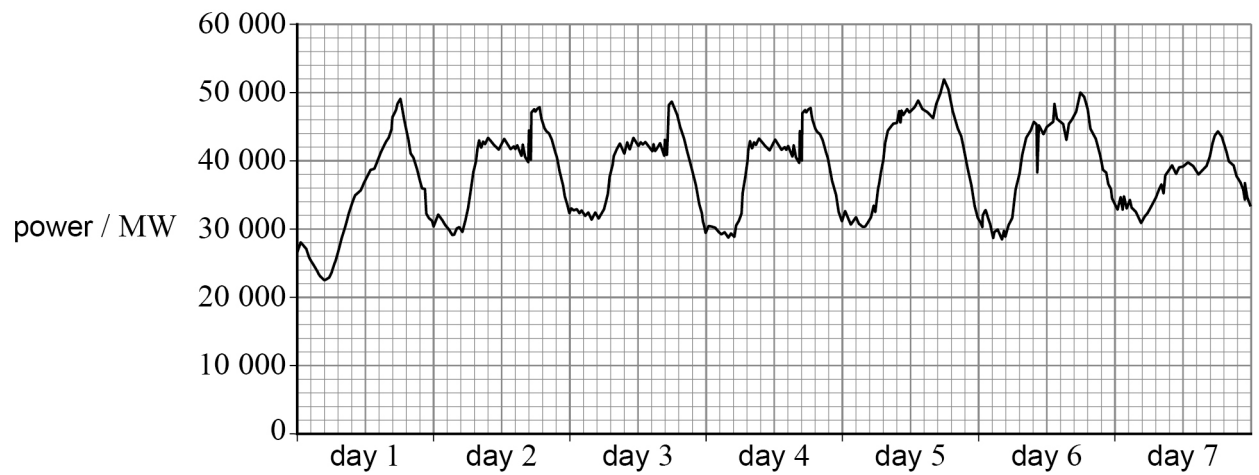
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0 5

A network of power stations and other power sources provides electrical power to industrial and residential consumers.

Figure 8 shows the variation of total power provided by the power stations and other power sources over a period of seven days.

Figure 8



0 5 . 1

Estimate, in J, the energy provided during the seven days shown in **Figure 8**. Give your answer to the nearest order of magnitude.

[3 marks]

order of magnitude of energy provided = _____ J

Some of the power was supplied from base-power stations and some was provided from back-up power stations. Power was also provided from other renewable sources.

0 5 . 2

Suggest a value for the power provided by base-power stations during the seven days shown in **Figure 8**.

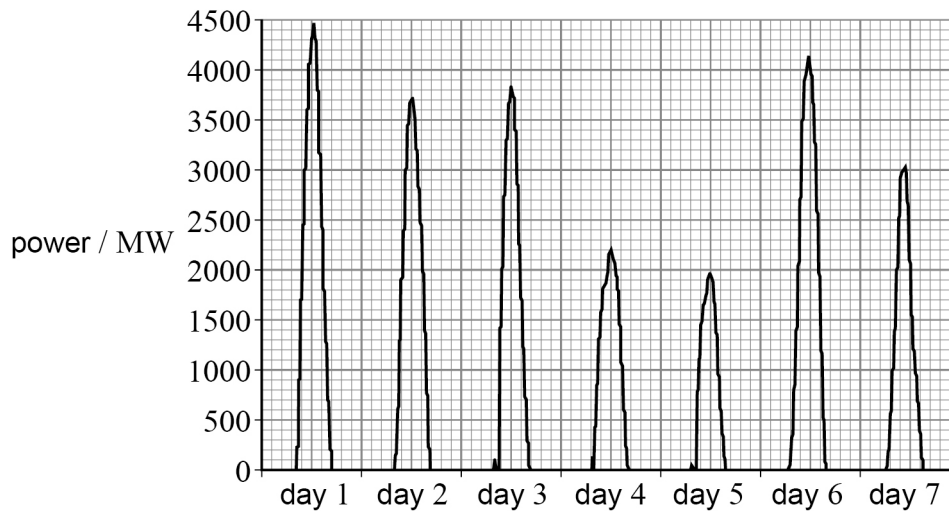
[1 mark]

power = _____ W



Figure 9 shows the contribution from solar power to the data in **Figure 8**.

Figure 9



0 5 . 3

Explain **one** similarity and **one** difference between the peaks in **Figure 9**.

[2 marks]

similarity _____

difference _____

0 5 . 4

The demands for power by consumers at noon on day 2 and day 4 were similar.

Explain how the contributions from solar power affected the power required from base-power stations and back-up power stations at these two times.

[3 marks]

base-power stations _____

back-up power stations _____

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A wind turbine has blades that sweep out a circular area of radius r .
The wind speed is v .
The maximum power available to a wind turbine is P .

Which combination of r and v produces the greatest value of P ?

[1 mark]

	r / m	$v / \text{m s}^{-1}$	
A	7	15	☐
B	8	12	☐
C	9	9	☐
D	10	6	☐

1 6 A satellite is 4.5×10^{10} m from the Sun.

The solar cells on the satellite have an area of 0.5 m^2 and provide 240 W to power the satellite when the solar cells face the Sun.

What percentage of the total solar power incident on the solar cells is used to provide power to the satellite?

power output of the Sun = 3.8×10^{26} W

[1 mark]

A 0.4%

☐

B 0.8%

☐

C 1.6%

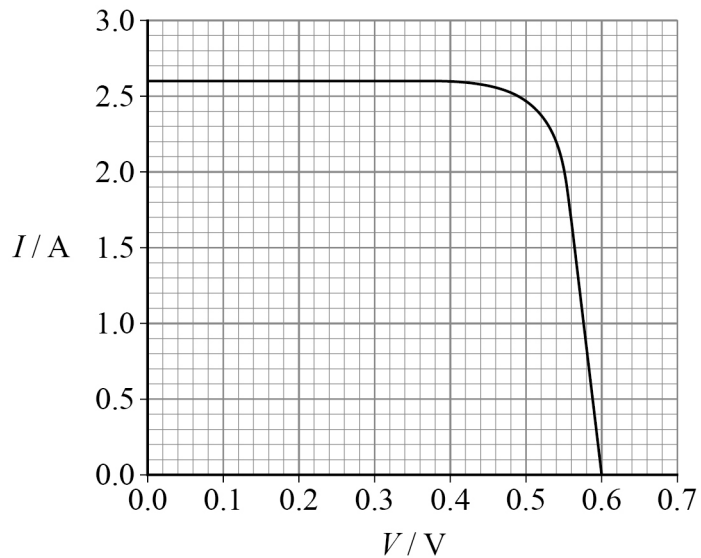
☐

D 3.2%

☐

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The graph shows the I - V characteristic for a solar cell.



What is the maximum power for the solar cell?

[1 mark]

A 1.04 W

☐

B 1.20 W

☐

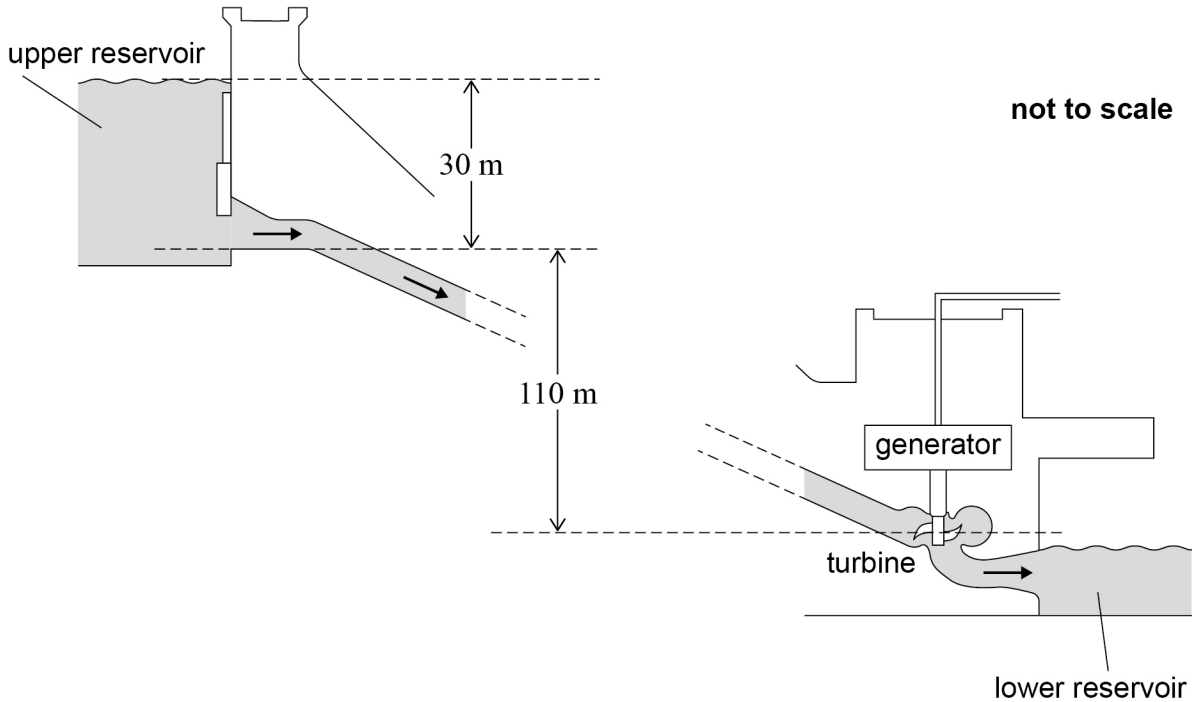
C 1.23 W

☐

D 1.56 W

☐

1 9 Water flows through the turbine of a hydroelectric power station at a rate of $250 \text{ m}^3 \text{ s}^{-1}$. The height of the outflow of the upper reservoir is 110 m above the turbine. The depth of the water in the upper reservoir is 30 m. The density of water is 1000 kg m^{-3} .



What is the maximum power available from the flow of water through the turbine?

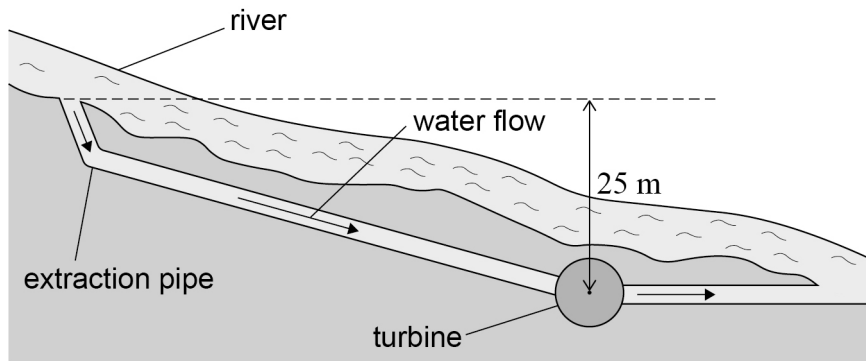
[1 mark]

- A** 74 MW ☐
- B** 270 MW ☐
- C** 310 MW ☐
- D** 340 MW ☐

Figure 1 shows part of a small hydroelectric power (HEP) station. Some of the river water enters the extraction pipe. This water flows through a turbine before it re-enters the river.

The flow in the river can vary, depending on the weather.

Figure 1



For this HEP station:

- the water falls a vertical distance of 25 m
- the mean electrical power output is 79 kW
- the efficiency is 86%.

Calculate the mean mass of water entering the extraction pipe every second.

[3 marks]

mean mass = _____ kg



0 1 . 2

The HEP station has an alternating output of voltage 69 kV rms.

Calculate the peak current output from the HEP station.

[2 marks]

peak current = _____ A

A second type of HEP station uses a dam and a reservoir to store water.

0 1 . 3

Suggest **one** environmental benefit of the station shown in **Figure 1** compared with one that uses a dam and a reservoir.

[1 mark]

0 1 . 4

State what is meant by a base-power station.

Go on to discuss whether either type of HEP station can operate as a base-power station.

[3 marks]



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A remote island has renewable sources that generate between 2300 kW and 2900 kW of electrical power.

The island needs a continuous power supply of 2500 kW.

The island also has a pumped storage system that stores an energy of 60 MW h.

The output of the renewable resources falls to its minimum value and remains at this minimum value.

The renewable sources together with the pumped storage system can only supply the island's needs for a time t .

What is t ?

[1 mark]

A 300 h

☐

B 150 h

☐

C 30 h

☐

D 15 h

☐

Don't write

A wind farm is being designed to provide power to a small town.

0	6
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1

Explain the factors that affect the design of the wind farm and its location.

In your answer you should consider:

- the factors that affect the maximum power that is available from a wind turbine
- the factors that determine the number and arrangement of the turbines
- the environmental factors involved in the use of wind turbines.

[6 marks]

[illegible]

Turn over ►



A solar farm is suggested as an alternative to the wind farm.

The solar farm consists of an array of solar panels. Each panel is angled so that the average intensity of the sunlight on the panels is at a maximum.
Each panel has an area of 1.0 m^2 .

The power per square metre from the Sun at a distance equal to the orbital radius of the Earth is 1.34 kW m^{-2} .

0 6 . 2 Calculate the power output of the Sun.

radius of Earth's orbit = 150 Gm

[2 marks]

power output = _____ W

0 6 . 3 The maximum output of a 1.0 m^2 solar panel is typically about 250 W in bright conditions.

Discuss why this is significantly smaller than 1.34 kW .

[3 marks]

END OF SECTION A

Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.

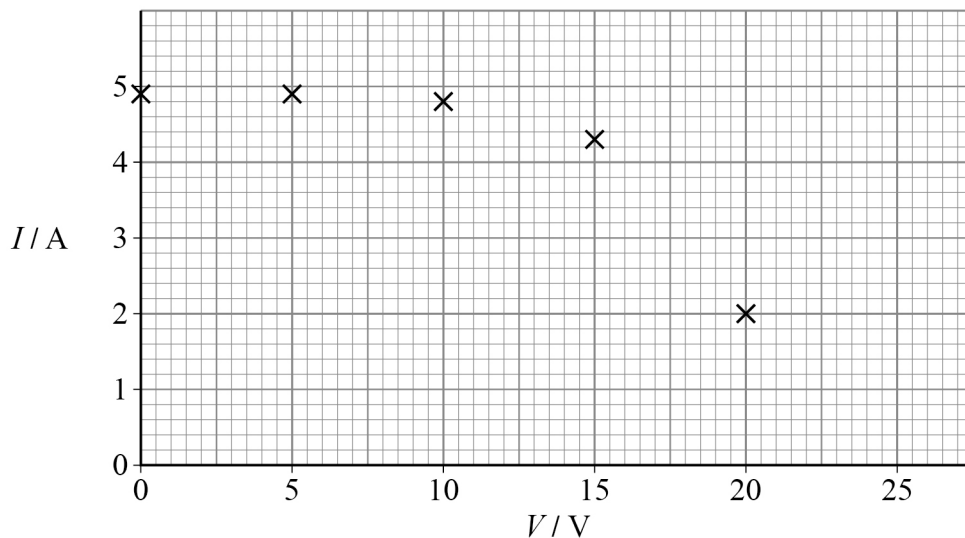


If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

A student takes measurements to obtain an I – V characteristic for an array of solar panels. The diagram shows the student's results.



The student decides to take more readings to obtain an accurate value for the maximum power output.

What range of voltages should the student investigate?

[1 mark]

A 0 to 6 V

☐

B 6 V to 12 V

☐

C 12 V to 18 V

☐

D 18 V to 24 V

☐